

GAN-Based Leaf Image Inpainting for Occlusion Recovery

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Abstract— *This study explores a novel Generative Adversarial Network (GAN) based pipeline for restoring occluded leaf images in the context of precision agriculture. This study investigates three approaches: an occlusion detection module built upon an autoencoder architecture, a U-Net-based generator for semantic inpainting and a PatchGAN discriminator designed to enhance perceptual realism. These approaches were thoroughly evaluated using a synthetically occluded leaf image dataset, yielding significant improvements in the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The results achieved an average PSNR of 23.16 dB and an SSIM of 0.6296, demonstrating the effectiveness of the model in restoring fine details and enhancing image quality. Among the three approaches, the Hybrid GAN outperforms the Standard GAN and Attention GAN, showing the best performance in both PSNR and SSIM. The findings further demonstrate the potential of the proposed system to support downstream agricultural applications such as automated plant disease detection, growth stage monitoring, and intelligent decision-making in smart farming systems.*

Keywords—Generative Adversarial Networks (GANs), Leaf Inpainting, Precision Agriculture, Plant Disease Detection, Precision Farming

I. INTRODUCTION

In recent years, integrating artificial intelligence (AI) into agricultural systems has revolutionized how crop health and productivity are monitored, primarily through computer vision and deep learning techniques. Among these, leaf imagery has emerged as a powerful tool for identifying diseases, estimating yield, and assessing plant vitality. However, a persistent challenge in this domain is the presence of occlusions—caused by overlapping leaves, pest infestations, environmental factors such as shadows and water droplets, or even equipment limitations—that significantly degrade image quality and reduce the reliability of automated analyses [11], [12].

Occlusion not only obstructs key features in the image but also introduces noise that can mislead classification and detection models, ultimately affecting downstream decision-making processes. Traditional image preprocessing techniques and classical inpainting algorithms have shown limited success in handling these occlusions, particularly in preserving semantic and structural continuity in complex agricultural scenes [6], [13]. To overcome these limitations, the research community has increasingly turned to Generative Adversarial Networks (GANs) [1], which have demonstrated impressive capabilities in image inpainting, especially in fields such as medical imaging [10] and natural scene restoration [4], [5].

Despite their potential, existing GAN-based inpainting models often struggle to generalize to agricultural datasets, where occlusions are context-dependent and spatially diverse. Moreover, many models fail to incorporate environmental semantics, resulting in unrealistic reconstructions that do not align with actual leaf physiology or crop conditions. In response to these limitations, we propose a robust GAN-based framework designed explicitly for occlusion recovery in leaf images. Our approach integrates a U-Net-based generator [2] for spatial detail preservation, a PatchGAN discriminator [3] for localized texture evaluation, and an autoencoder-based occlusion detection module [7] for precise mask generation.

In this study, we explore and compare three variations of our framework: a baseline GAN for leaf image inpainting, an attention-augmented GAN for improved focus on occluded regions [8] and a perceptual loss-enhanced hybrid GAN for better performance in terms of image quality. Our experiments demonstrate that the third model significantly outperforms the others in terms of both quantitative metrics and qualitative realism.

This research contributes to developing intelligent, context-aware agricultural monitoring systems that can operate effectively under real-world conditions, such as those encountered in the diverse agrarian landscapes of Sri Lanka. By enhancing the reliability of image-based crop analysis, the proposed method can support early disease detection, improve yield forecasting, and promote sustainable farming practices.

II. RELATED WORK

Image inpainting has evolved significantly, with Generative Adversarial Networks (GANs) becoming the state-of-the-art method for generating realistic inpainted images. Goodfellow et al. [1] introduced GANs as a framework where the Generator creates images, and the Discriminator distinguishes between real and fake, leading to improved image quality. U-Net [2] architecture revolutionized image segmentation and inpainting by using an encoder-decoder structure, making it well-suited for generating images with high spatial resolution.

The conditional GANs (cGANs) [3] enabled more controlled image-to-image translation tasks, like image inpainting, by conditioning the Generator on the input image. More recent work has focused on incorporating attention mechanisms to help the generator focus on occluded regions during inpainting, significantly improving context awareness and realism. Chen and Liu [8] successfully applied attention-guided inpainting to improve image completion by allowing the Generator to focus on specific areas of the image.

In agricultural contexts, GANs have been used for leaf image reconstruction [9], [11], where the challenge is not just to restore the missing parts but also to preserve the texture and fine details of leaves, which are essential for accurate plant health analysis. Yu et al. [9] and Xie et al. [11] demonstrated that GAN-based models can effectively restore missing or occluded leaf parts, making them useful for agricultural monitoring.

Despite the remarkable progress in GAN-based image inpainting, several limitations persist when these methods are applied to agricultural imaging contexts.

Existing approaches do not explicitly detect or localize occlusions before inpainting, which can lead to overgeneralized or contextually irrelevant reconstructions. Additionally, while attention mechanisms and perceptual losses have been employed individually to enhance realism, the synergistic integration of these techniques alongside environmental context has not been comprehensively explored. These gaps underscore the need for a specialized, end-to-end framework that detects and localizes occlusions and leverages semantics to deliver contextually accurate restorations. This motivates the present study, which proposes and evaluates a GAN-based pipeline designed explicitly for leaf image inpainting in precision agriculture.

III. PROPOSED METHODOLOGY

To address the challenge of occlusion in leaf imagery, we propose and evaluate three distinct approaches based on a unified GAN-based architecture. While all three methods share a common foundational framework, their internal mechanisms and enhancement strategies differ for achieving robust and context-aware image inpainting.

The Generator is based on the U-Net architecture, a well-known model for image-to-image tasks like inpainting. As shown in Fig.1, the network follows a typical encoder-decoder structure with skip connections, which help preserve fine details during image reconstruction. The input is a $256 \times 256 \times 3$ RGB leaf image with occluding regions. In the encoder, the image is progressively downsampled to extract high-level features. The decoder then up-samples the encoded features to reconstruct the occluded regions. Finally, the output is a fully reconstructed image with the occluded regions inpainted.

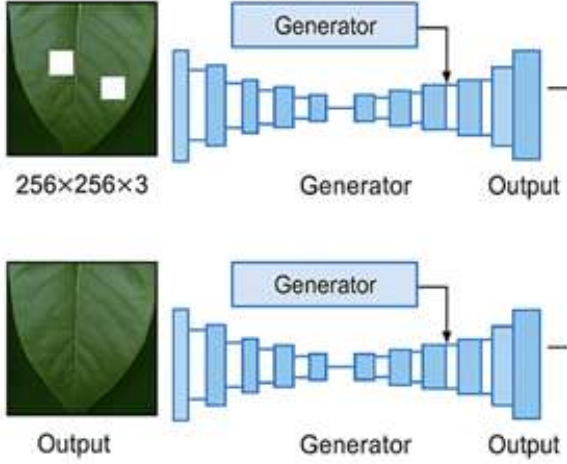


Fig. 1 Standard GAN-based Image Inpainting

The generator network can be mathematically represented during the down-sampling and up-sampling steps, as shown in Equation (1).

$$E(x) = f(W_E \cdot x + b_E) \quad (1)$$

Where $E(x)$ is the output of the encoder, W_E is the weight of encoder, b_E is bias of encoder, x is the input image, and f is the activation function applied during down-sampling.

Afterwards, the decoder performs the up-sampling of the encoded feature maps, as shown in Equation (2).

$$D(E(x)) = g(W_D \cdot E(x) + b_D) \quad (2)$$

Where $D(E(x))$ represents the decoder output, W_D is the weight of the decoder, b_D is bias of the decoder, and g is the activation function used during up-sampling.

The Discriminator used in our approach is a PatchGAN, which evaluates local patches of the image rather than the entire image. This approach enforces realism in the reconstructed regions by focusing on high-frequency details and ensuring that the inpainted areas blend seamlessly with the rest of the image. The Discriminator receives both real images and generated inpainted images as input. Its output is a binary classification for each patch, distinguishing between real and generated patches.

As in Approach One, a basic GAN architecture is implemented for image inpainting, where the Generator is trained to reconstruct the occluded regions in the image

while the Discriminator distinguishes real from fake regions. The Generator uses a U-Net-inspired architecture, and the Discriminator operates as a standard PatchGAN.

As Approach Two, we introduce attention mechanisms into the GAN architecture as shown in Fig. 2 to improve the Generator's ability to focus on the occluded regions during inpainting. The attention mechanism helps the Generator to selectively focus on the occluded regions, overcoming the challenge of global context awareness. The attention mechanism $A(Q, K, V)$ can be mathematically represented as equation 3.

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

Where Q , K^T , and V represent the query, transpose of key, and value matrices, respectively, and d_k is the

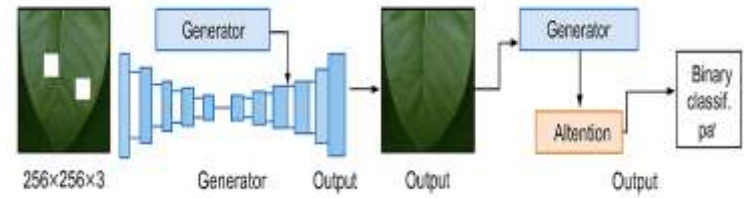


Fig. 2: GAN with Attention Mechanisms

dimension of the key vectors. This attention module allows the Generator to focus more effectively on missing or occluded areas.

In Approach Three, we integrate perceptual loss into the GAN framework to preserve texture consistency and enhance fine details in the inpainted areas, as shown in Fig.3. Perceptual loss helps the Generator focus on maintaining the natural texture and fine structures of the leaf, which improves the realism of the inpainted images. The perceptual loss can be formulated as Equation (4).

$$L_{\text{perceptual}} = \sum (\lambda_i \left\| \varphi_i(G(x)) - \varphi_i(x) \right\| 1) \quad (4)$$

Where φ_i represents the feature maps from the i -th layer of a pre-trained network (e.g., VGG16), and λ_i is the weighting factor for each layer's contribution to the loss. This perceptual loss helps ensure that the inpainted regions not only match the surrounding context but also preserve the finer textures and structures inherent in the original image.

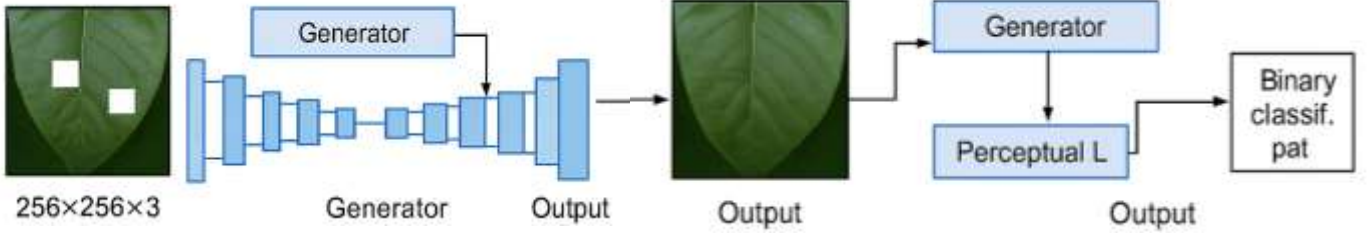


Fig. 3 : Hybrid GAN with Perceptual Loss

IV. DATASET AND PREPROCESSING

The dataset used in this study was sourced from the PlantVillage dataset, which contains high-resolution leaf images of various plant species. To simulate real-world scenarios where portions of the leaf might be missing or damaged due to factors such as disease, environmental damage, or occlusions, we created realistic occlusions in the images. Specifically, occlusions were introduced by masking random regions of the leaf images with black rectangles, representing tears, holes, and partial visibility. Each original leaf image was then augmented to generate multiple occluded variants, allowing the model to learn to recover the missing regions.

To ensure compatibility with the model architecture, all images—both clean and occluded variants—were resized to 256×256 pixels. This resizing step ensured uniformity across the dataset and met the input size requirements of the deep learning model. Additionally, all pixel values were normalized to a range of $[0, 1]$ by dividing each pixel by 255, preparing the data for efficient model training. The occlusion masks generated during preprocessing were used to guide the Generator during training, ensuring that only the missing regions were inpainted. The masks were multiplied with the generated output to combine the restored region with the unoccluded parts of the image.

The dataset was split into 80% for training, 10% for validation, and 10% for testing. This partitioning allows for effective training and evaluation of the model's performance.

V. EXPERIMENTAL RESULTS

The performance of the models was evaluated using PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index), and visual inspection to assess their

ability to reconstruct occluded regions in the leaf images. PSNR was used to measure the pixel-level differences between the inpainted and original clean images. Higher PSNR values indicate a better reconstruction with fewer pixel differences.

SSIM was used to assess the structural similarity between the original and inpainted images. An SSIM value closer to 1.0 indicates a better match in structural information between the images.

Visual inspection facilitated a qualitative assessment of the generated images, with particular attention to the reconstruction of fine details such as leaf venation and edge continuity.

Standard GAN-based Image Inpainting approach provided acceptable inpainting results, as shown in Table 1. While it was able to recover some of the occluded regions, it struggled to preserve fine leaf veins and sharp edges. The PSNR and SSIM values were moderate (PSNR: 13.48 dB, SSIM: 0.2004), indicating a reasonable but not optimal recovery of missing areas. Despite performing decently, the Standard GAN showed limitations in restoring finer structural details, such as leaf veins, resulting in lower-quality reconstructions than more advanced models. These moderate values suggest that the model's ability to handle complex textures and fine details remains limited.

GAN with Attention Mechanisms helped the Generator focus on the occluded regions more effectively. As a result, the model produced slightly better results than the standard GAN, especially in terms of maintaining leaf vein structures and improving edge continuity. The PSNR and SSIM scores showed improvement compared to the previous, but the results were still not perfect, and there is still room for enhancement, particularly in handling more complex textures and finer details.

Hybrid GAN with Perceptual Loss model integrated perceptual loss with attention mechanisms performed the best in preserving texture and fine details. Although the model performed better than the other two approaches (PSNR: 23.16 dB, SSIM: 0.6296), indicating better restoration of both pixel-level and structural details. However, despite these advancements, further refinement is needed in terms of texture consistency and fine-grained details in some areas, particularly in more complex textures such as leaf veins and edges.

Table 1: Performance Comparison of Inpainting Algorithms

<i>Approach</i>	<i>PSNR (dB)</i>	<i>SSIM</i>
Standard GAN-based Image Inpainting	13.48	0.2004
GAN with Attention Mechanisms	18.09	0.3144
Hybrid GAN with Perceptual Loss	23.16	0.6296

In summary, the attention-based GAN and the hybrid model outperformed the standard GAN in most aspects of image inpainting, particularly in recovering fine details. However, the overall performance still has limitations, especially in regions requiring more complex textures and finer details.

VI. CONCLUSION

This study investigated three GAN-based approaches for inpainting occluded leaf images to enhance visual realism, structural detail preservation, and contextual understanding in reconstructed regions. The proposed architecture was anchored by a U-Net-inspired generator designed for accurate localization and semantic recovery, supported by skip connections to preserve fine spatial features. A PatchGAN discriminator was employed to assess the local texture quality of inpainted regions, thereby enforcing high-frequency fidelity in output images.

Based on a standard GAN framework, the first approach demonstrated baseline competence in image completion but exhibited limitations in handling complex occlusions, particularly concerning texture consistency and edge sharpness. The second approach, incorporating attention mechanisms, significantly improved the model's ability

to selectively focus on occluded areas, resulting in more context-aware reconstructions. The third and most effective approach integrated perceptual loss derived from pre-trained feature extractors (e.g., VGG16), which enhanced semantic alignment and texture realism, yielding outputs that were perceptually closer to natural leaf structures.

The results underscore the benefits of progressively integrating attention mechanisms and perceptual cues into GAN-based inpainting pipelines. Quantitative metrics and qualitative assessments consistently indicated that the hybrid GAN with perceptual loss achieved the most robust performance, particularly in scenarios involving fine-grained occlusions and structurally complex leaf patterns.

The significance of this research lies in its direct applicability to real-world agricultural systems, where occlusion from overlapping foliage, pests, and environmental interference often hampers automated crop monitoring. The proposed models offer a promising solution for enhancing the accuracy of downstream tasks such as plant disease detection, phenotypic analysis, and yield estimation. In particular, the methodology is well-suited for deployment in data-scarce and occlusion-prone environments, such as the diverse and dynamic agricultural landscapes of Sri Lanka.

Future work will expand the dataset to include real-world occluded leaf imagery captured under varying lighting and climatic conditions. Additionally, we plan to explore the integration of temporal and multispectral data, enabling the development of a more comprehensive multimodal agricultural monitoring system. Finally, deploying edge devices for real-time field analysis and incorporating active learning strategies for continual model refinement are envisioned to support scalable and sustainable smart farming practices.

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